Speaking App PRD — Student Form

- **Name / Student ID:** 정승교 / 2022315564
- **My situation (상황):** Choosing dinner menu with friends

1. App Name (앱 이름)

- Whats For Dinner? - Dinner choices

2. Why We Are Building It (왜 만드는가)

- Problem: Students often struggle to come to a conclusion amongst themselves after discussions especially in a foreign language.
- Solution: Students Listen -> Choose Options -> Speak Target Sentence -> Get Feedback -> Negotiate -> Make decision
- Goal: Students can state their dinner preferences and explain their reasons with logical arguments

3. Target Users (대상 사용자)

- Grade: Middle school 2nd year
- Level: CEFR A2
- Characteristic: Shy about speaking aloud of their opinions due to being afraid of disagreements by others
- Common mistake: Opinion is slow to be transferred into speech, leading to incomplete or poorly worded arguments where speech lags behind thought.
- Use context: At the end of class or after school, 5–10 minutes, on a smartphone

4. Screen Layout (화면 구성)

Screen 1: Start Screen

- Top: App title "Whats For Dinner?" with subtitle "Dinner with friends"
- Middle: Middle: Illustration of 3 friends walking home after school, talking about food
- Bottom: Large start talking button. RIght at the bottom should be a Small privacy notice that says no recording will be kept

Screen 2: Practice Screen

- Top: Progress indicator (Step X/5) + Partner character speech bubble (e.g., "What do you want to eat tonight?")
- Middle: Option selector: 3 food visual cards with titles (e.g., Pizza, Tteokbokki, Burger). Reason tag selector: Appears after food choice (e.g., Spicy, Cheesy, Hot, Cheap, CHewy). Target sentence area: Dynamically generated sentence based on selections (largest text, e.g., "I want to eat Tteokbokki because it is spicy.") + listen button
- Bottom: Large round mic button [🎤 Tap to Speak] with state indicators (Ready: Grey → Listening: Green → Processing: Blue). Feedback card (Appears after speaking with character avatar). Action buttons: [Try Again] [Next Phase →]
- Mic button states: off/grey on/green

Screen 3: Result Screen

- Top: Celebration graphic + "Dinner Choice Finalized!" header
- Middle: Summary list showing the student's 5 recorded responses + Star performance badge
- Bottom: [Practice Again] button

5. Key Features (주요 기능)

1. See the situation — The partner character ask main question. The student then selects visual option cards and reasons to build their target sentence.
2. Listen — Student selects the listen button to hear the pronunciation of the created target sentence.
3. Speak — Pressing the mic activates recording. Recording automatically stops after 3 seconds of silence or mic is pressed again.
4. Compare — The app converts student speech to text using Web Speech API and compares key words to target setence
5. Feedback — The character gives short, supportive feedback based on target key words and feedback rules (recast, elicitation, or chunking) as required by the professor
6. Try again / Next — Student can retry the current turn or proceed to the next turn.

After all sentences are finished, the app moves to the Result Screen.

6. Practice Dialogue (연습 대화)

- **Partner character (name / role):** Random name / Friend

No.	Partner	Student target sentence	Korean meaning	Key words
1	What do you want to eat?	I want to eat pizza tonight.	나 오늘 저녁으로 피자 먹고 싶어.	want, pizza
2	Oh, pizza? Why do you want to eat that?	I want to eat pizza because it is chewy.	피자가 쫄깃해서 먹고 싶어.	pizza, because, delicious
3	But I ate pizza yesterday! How about something hot and spicy?	Then how about eating tteokbokki instead?	그럼 대신에 떡볶이 먹는 건 어때?	how about, tteokbokki
4	Tteokbokki sounds good, but is there a good place near here?	Yes, there is a famous shop near our school.	응, 우리 학교 근처에 유명한 가게가 있어.	famous, shop, near
5	Perfect! Should we go there?	Yeah let's go eat there	좋아 지금 가자.	Let's, go,

- **Star rules (reward completion, not accuracy):**
  - ⭐⭐⭐
  - ⭐⭐
  - ⭐ Finished the whole dialogue

7. Feedback Rules (피드백 규칙)

- **Character personality / speaking style:** Friendly classmate, uses simple grammar. Different to feedback providier
- **My rules:**
  1. Always praise first before giving constructive tips.
  2. Limit feedback to two sentences max (English sentence + Korean translation)
  3. Minimise use of negative words and percentage scores
  4. Judge based on communication level. Was the main idea sent across?
  5. Follow the order of 1st fail (Recast) → 2nd fail (Elicitation with first letter) → 3rd fail (Chunking).

Situation	Feedback (English + Korean)
Success	Great job I understood your point. (잘했어 네가 뭐라고 말하는지 이해했어.)
Success + small word missing	Good job just remember these words next time. (잘했는데 다음엔 이 단어들을 잊지마.)
1st failure (recast)	Nice try but this time can you say: "I want to eat pizza because..." (잘했는데 이번에는 이렇게 말해 볼까?)
2nd failure (elicitation)	Nice try but when reasoning you need to use a word that starts with B. "I want to eat pizza b___ it is delicious." ('이유'를 말할 땐 b로 시작하는 단어가 필요해.)
3rd failure (chunking)	Let's try saying it in short chunks. Repeat after me: "because it is delicious." (이번에는 짧게 나눠서 따라 말해보자)
No speech recognized	I couldn't hear you, could you tap the mic and try speaking again? (소리가 안들렸어. 마이크를 누르고 다시 말해 주겠어?)

8. Do Not (하지 말 것)

Privacy and Safety (required — keep these)

- Do not save or send audio recordings.
- Do not collect personal information (name, student ID, school, email). No login or sign-up.
- No ads, external links, or payment features.
- No photos of real people or famous characters.

Learning Design (add at least 2)

- Do not use point systems
- Do not allow students to skip to the next phase without creating their target sentence
- Do not focus on pronounciation. That should be done in real life.

Other (optional)

- Do not use complex grammatical structures and rare phrases

9. Completion Checklist (완성 체크리스트)

- [ ] The mic button visibly changes colour (Grey -> Green) when tapped.
- [ ] What the student said appears as text in real time.
- [ ] Pronounciation errors (that are not word altering) are not counted as failures.
- [ ] After all sentences are completed, the Result Screen appears.
- [ ] Target sentence dynamically updates on Screen 2 when the user selects different food items and reason tags.
- [ ] Feedback follows the Recast → Elicitation → Chunking repair strategy on consecutive failures.

Technical Requirements (do not edit — 수정하지 말고 그대로 두세요)

- **Speech recognition:** Browser built-in Web Speech API (SpeechRecognition), language `en-US`
- **Read-aloud:** Browser built-in SpeechSynthesis, language `en-US`, normal rate 0.9 / slow rate 0.7
- **Feedback sentences:** Gemini, following Section 7. If Gemini does not respond, show the matching example sentence from the Section 7 table. The app must work to the end without AI.
- Do **not** run SpeechRecognition and MediaRecorder at the same time. Use speech recognition only.
- Disable the mic button while the model audio is playing.
- Show a guidance message when: the browser does not support speech recognition ("Please open it in Chrome."), microphone permission is denied, or the app is opened in an in-app browser such as KakaoTalk or Instagram.
- Show a small notice on the Start Screen: "Your voice is processed to turn it into text and is not recorded or saved."
- The deployed address must use **https** for the microphone to work.

AI use log

- AI used to create app Screen Layout creation based on rest of my PRD
- AI used to polish PRD structure making sure there are no overlaps

Before You Submit (제출 전 자기 점검)

- [ ] Section 3 includes a **CEFR level**, and the sentences in Section 6 match it.
- [ ] Section 4 uses **Top / Middle / Bottom** and matches my Stitch screens.
- [ ] Section 6 has **5 turns** with key words.
- [ ] Section 7 has **recast → elicitation → chunking** and no % scores.
- [ ] Section 9 has sentences I can check by testing.
- [ ] My partner checked this PRD, and I revised it.
- [ ] I recorded the AI tools and main prompts in my AI use log (.md).